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科目: 概率論與隨機過程

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Problem 9.

$$\begin{aligned}
 1. \quad f_{X+Y}(z) &= \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x) dx = \int_{-\infty}^{+\infty} f_X(x) f_Y(x-z) dx \\
 &= \int_{-\infty}^{+\infty} f_X(x) f_Y(a+b-(x-z)) dx \\
 &= \int_{-\infty}^{+\infty} f_X(x) f_Y(a+b+z-x) dx = f_{X+Y}(a+b+z)
 \end{aligned}$$

$$2. \quad \text{var}(R) = \text{var}(W+X) = \text{var}(W) + \text{var}(X) = 2$$

$$\text{var}(S) = \text{var}(X+Y) = \text{var}(X) + \text{var}(Y) = 2$$

$$\text{var}(T) = \text{var}(Y+Z) = \text{var}(Y) + \text{var}(Z) = 2$$

$$\begin{aligned}
 \text{cov}(R, S) &= \text{cov}(W+X, X+Y) = E[(W+X)(X+Y)] - E[W+X]E[X+Y] \\
 &= E[W]E[X] + E[W]E[Y] + E[X]E[Y] + E[X^2] - (E[W] + E[X])(E[X] + E[Y]) \\
 &= E[X^2] = \text{var}(X) + (E[X])^2 = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{cov}(R, T) &= \text{cov}(W+X, Y+Z) = E[(W+X)(Y+Z)] - E[W+X]E[Y+Z] \\
 &= E[W]E[Y] + E[W]E[Z] + E[X]E[Z] + E[X]E[Y] - 0 \\
 &= 0
 \end{aligned}$$

$$\rho(R, S) = \frac{\text{cov}(R, S)}{\sqrt{\text{var}(R)\text{var}(S)}} = \frac{1}{2}, \quad \rho(R, T) = \frac{\text{cov}(R, T)}{\sqrt{\text{var}(R)\text{var}(T)}} = 0$$

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3. $f_{X_r}(z) = \lambda_r e^{-\lambda_r z} \quad (r=1, 2, \dots)$

$$\begin{aligned} f_{X_r+X_s}(z) &= \int_0^z f_{X_s}(x) f_{X_r}(z-x) dx = \int_0^z \lambda_s \lambda_r e^{-\lambda_s x} e^{-\lambda_r(z-x)} dx \\ &= \frac{\lambda_s \lambda_r}{\lambda_r - \lambda_s} e^{-\lambda_r z} (e^{(\lambda_r - \lambda_s)z} - 1) = \frac{\lambda_s \lambda_r}{\lambda_r - \lambda_s} (e^{-\lambda_s z} - e^{-\lambda_r z}) \\ &= \frac{\lambda_r}{\lambda_r - \lambda_s} f_{X_s}(z) + \frac{\lambda_s}{\lambda_s - \lambda_r} f_{X_r}(z) \end{aligned}$$

$$\Rightarrow f_{X_1+X_2+\dots+X_n}(z) = \sum_{k=1}^n \left(\prod_{\substack{1 \leq t \leq n \\ t \neq k}} \frac{\lambda_t}{\lambda_t - \lambda_k} \right) \lambda_k e^{-\lambda_k z} \quad [\text{Induction}]$$

When $n=1$, L.H.S. = $f_{X_1}(z) = \lambda_1 e^{-\lambda_1 z} = \text{R.H.S.}$

Assume ~~$n=k$~~ the statement is correct when $n=k$ ($k \geq 1$),

then when $n=k+1$

$$\text{L.H.S.} = f_{X_1+X_2+\dots+X_k+X_{k+1}}(z) = \int_0^z f_{X_1+X_2+\dots+X_k}(x) f_{X_{k+1}}(z-x) dx$$

$$= \sum_{t=1}^k \left(\prod_{\substack{1 \leq r \leq k \\ r \neq t}} \frac{\lambda_r}{\lambda_r - \lambda_t} \right) \lambda_t \int_0^z f_{X_t}(x) f_{X_{k+1}}(z-x) dx$$

$$= \sum_{t=1}^k \left(\prod_{\substack{1 \leq r \leq k \\ r \neq t}} \frac{\lambda_r}{\lambda_r - \lambda_t} \right) \left(\frac{\lambda_t}{\lambda_t - \lambda_{k+1}} f_{X_{k+1}}(z) + \frac{\lambda_{k+1}}{\lambda_{k+1} - \lambda_t} f_{X_t}(z) \right)$$

$$= \sum_{t=1}^k \left(\prod_{\substack{1 \leq r \leq k+1 \\ r \neq t}} \frac{\lambda_r}{\lambda_r - \lambda_t} \right) f_{X_t}(z) + \sum_{t=1}^k \left(\prod_{\substack{1 \leq r \leq k \\ r \neq t}} \frac{\lambda_r}{\lambda_r - \lambda_t} \right) \frac{\lambda_t}{\lambda_t - \lambda_{k+1}} f_{X_{k+1}}(z)$$

$$= \sum_{t=1}^k \left(\prod_{\substack{1 \leq r \leq k+1 \\ r \neq t}} \frac{\lambda_r}{\lambda_r - \lambda_t} \right) f_{X_t}(z) + \sum_{t=1}^k \left(\prod_{\substack{1 \leq r \leq k+1 \\ r \neq t}} \frac{\lambda_r}{\lambda_r - \lambda_t} \right) \frac{\lambda_t}{\lambda_t - \lambda_{k+1}} f_{X_{k+1}}(z)$$

$$= \sum_{t=1}^k \left(\prod_{\substack{1 \leq r \leq k+1 \\ r \neq t}} \frac{\lambda_r}{\lambda_r - \lambda_t} \right) f_{X_t}(z) + \left(\prod_{\substack{1 \leq r \leq k+1 \\ r \neq k+1}} \frac{\lambda_r}{\lambda_r - \lambda_{k+1}} \right) f_{X_{k+1}}(z)$$

$$= \sum_{t=1}^{k+1} \left(\prod_{\substack{1 \leq r \leq k+1 \\ r \neq t}} \frac{\lambda_r}{\lambda_r - \lambda_t} \right) f_{X_t}(z) = \sum_{t=1}^{k+1} \left(\prod_{\substack{1 \leq r \leq k+1 \\ r \neq t}} \frac{\lambda_r}{\lambda_r - \lambda_t} \right) \lambda_t e^{-\lambda_t z} = \text{R.H.S.}$$

$$4. \text{ (a) } f_{X,Y}(x,y) = \frac{1}{\sigma \tau 2\pi \sqrt{1-\rho^2}} \exp \left[- \left(\left(\frac{x}{\sigma} \right)^2 + \left(\frac{y}{\tau} \right)^2 - 2\rho \left(\frac{x}{\sigma} \right) \left(\frac{y}{\tau} \right) \right) / 2(1-\rho^2) \right]$$

$$= \frac{\sqrt{1-\rho^2}}{\sigma \sqrt{1-\rho^2} \cdot \sqrt{2\pi} \cdot \tau \sqrt{1-\rho^2} \cdot \sqrt{2\pi}} \cdot \exp \left[- \left[\frac{x^2}{2(\sigma \sqrt{1-\rho^2})^2} + \frac{y^2}{2(\tau \sqrt{1-\rho^2})^2} - \frac{2\rho x y}{\sigma \sqrt{1-\rho^2} \cdot \tau \sqrt{1-\rho^2}} \right] \right]$$

$$= \frac{\sqrt{1-\rho^2}}{\sigma \sqrt{1-\rho^2} \cdot \sqrt{2\pi} \cdot \tau \sqrt{1-\rho^2} \cdot \sqrt{2\pi}} \exp \left[- \left[\frac{1}{2} \left(\frac{x}{\sigma \sqrt{1-\rho^2}} - \rho \frac{y}{\tau \sqrt{1-\rho^2}} \right)^2 + \frac{1}{2} \left(\frac{y}{\tau} \right)^2 \right] \right]$$

$$= \frac{\sqrt{1-\rho^2}}{\sigma \sqrt{1-\rho^2} \sqrt{2\pi} \cdot \tau \sqrt{1-\rho^2} \sqrt{2\pi}} \exp \left[- \left[\frac{1}{2} \left(\frac{y}{\tau \sqrt{1-\rho^2}} - \rho \frac{x}{\sigma \sqrt{1-\rho^2}} \right)^2 + \frac{1}{2} \left(\frac{x}{\sigma} \right)^2 \right] \right]$$

~~fix~~

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{1}{\sigma \sqrt{1-\rho^2} \cdot \sqrt{2\pi}} \exp \left[- \frac{1}{2} \left(\frac{x}{\sigma \sqrt{1-\rho^2}} - \rho \frac{y}{\tau \sqrt{1-\rho^2}} \right)^2 \right]$$

$$= \frac{1}{\sigma \sqrt{1-\rho^2} \cdot \sqrt{2\pi}} \exp \left[- \frac{(x - \frac{\rho \sigma}{\tau} y)^2}{2(\sigma \sqrt{1-\rho^2})^2} \right] = N \left(\underset{\uparrow}{\frac{\rho \sigma}{\tau} y}, \underset{\uparrow}{\sigma^2(1-\rho^2)} \right)$$

$$\Rightarrow (a) E[X|Y] = \frac{\rho \sigma}{\tau} Y, \quad (b) \text{var}(X|Y) = \sigma^2(1-\rho^2)$$

Denote that ρ

$$f_{X+Y}(z) = \int_{-\infty}^{+\infty} f_{X,Y}(x, z-x) dx = \frac{1}{2\pi\sigma\tau\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} \exp\left[-\frac{\left(\frac{1}{\sigma^2} + \frac{1}{\tau^2} + \frac{2\rho}{\sigma\tau}\right)x^2 - \left(\frac{2z}{\tau^2} + \frac{2\rho z}{\sigma\tau}\right)x + \left(\frac{z}{\tau}\right)^2}\right] dx$$

$$= \frac{1}{2\pi\sigma\tau\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} \exp\left[-\frac{(\sigma^2 + \tau^2 + 2\rho\sigma\tau)x^2 - 2(\sigma^2 + \rho\sigma\tau)zx + \sigma^2 z^2}{2(1-\rho^2)\sigma^2\tau^2}\right] dx$$

$$= \frac{1}{2\pi\sigma\tau\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} \exp\left[-\frac{x^2 - 2\left(\frac{\sigma^2 + \rho\sigma\tau}{\sigma^2 + 2\rho\sigma\tau + \tau^2}\right)zx + \frac{\sigma^2}{\sigma^2 + 2\rho\sigma\tau + \tau^2}z^2}{\frac{2(1-\rho^2)\sigma^2\tau^2}{\sigma^2 + 2\rho\sigma\tau + \tau^2}}\right] dx$$

$$= \frac{1}{2\pi\sigma\tau\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} \exp\left[-\frac{\left(x - \frac{\sigma^2 + \rho\sigma\tau}{\sigma^2 + 2\rho\sigma\tau + \tau^2}z\right)^2 + \frac{\sigma^4 + 2\rho\sigma^3\tau + \sigma^2\tau^2 - \sigma^4 - 2\rho\sigma^3\tau - \rho^2\sigma^2\tau^2}{(\sigma^2 + 2\rho\sigma\tau + \tau^2)^2}z^2}{2 \cdot \frac{(1-\rho^2)\sigma^2\tau^2}{\sigma^2 + 2\rho\sigma\tau + \tau^2}}\right] dx$$

$$= \frac{1}{2\pi\sigma\tau\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} \exp\left[-\frac{\left(x - \frac{\sigma^2 + \rho\sigma\tau}{\sigma^2 + 2\rho\sigma\tau + \tau^2}z\right)^2}{2 \cdot \frac{(1-\rho^2)\sigma^2\tau^2}{\sigma^2 + 2\rho\sigma\tau + \tau^2}}\right] \cdot \exp\left[-\frac{z^2}{2(\sigma^2 + 2\rho\sigma\tau + \tau^2)}\right] dx$$

$$= \frac{1}{2\pi\sigma\tau\sqrt{1-\rho^2}} \exp\left[-\frac{z^2}{2(\sigma^2 + 2\rho\sigma\tau + \tau^2)}\right] \cdot \sqrt{2\pi} \cdot \frac{\sqrt{1-\rho^2}}{\sqrt{\sigma^2 + 2\rho\sigma\tau + \tau^2}} \cdot \sigma\tau$$

$$= \frac{1}{\sqrt{\sigma^2 + 2\rho\sigma\tau + \tau^2} \cdot \sqrt{2\pi}} \exp\left[-\frac{z^2}{2(\sigma^2 + 2\rho\sigma\tau + \tau^2)}\right]$$

$$f_{X|X+Y}(x|z) = \frac{f_{X,Y}(x, z-x)}{f_{X+Y}(z)} = \frac{\frac{1}{2\pi\sigma\tau\sqrt{1-\rho^2}} \exp\left[-\frac{\left(x - \frac{\sigma^2 + \rho\sigma\tau}{\sigma^2 + 2\rho\sigma\tau + \tau^2}z\right)^2}{2 \cdot \frac{(1-\rho^2)\sigma^2\tau^2}{\sigma^2 + 2\rho\sigma\tau + \tau^2}}\right] \cdot \exp\left[-\frac{z^2}{2(\sigma^2 + 2\rho\sigma\tau + \tau^2)}\right]}{\frac{1}{\sqrt{\sigma^2 + 2\rho\sigma\tau + \tau^2} \cdot \sqrt{2\pi}} \exp\left[-\frac{z^2}{2(\sigma^2 + 2\rho\sigma\tau + \tau^2)}\right]}$$

$$= \frac{1}{\frac{\sqrt{1-\rho^2}\sigma\tau}{\sqrt{\sigma^2 + 2\rho\sigma\tau + \tau^2}} \cdot \sqrt{2\pi}} \exp\left[-\frac{\left(x - \frac{\sigma^2 + \rho\sigma\tau}{\sigma^2 + 2\rho\sigma\tau + \tau^2}z\right)^2}{2 \cdot \left(\frac{\sqrt{1-\rho^2}\sigma\tau}{\sqrt{\sigma^2 + 2\rho\sigma\tau + \tau^2}}\right)^2}\right] = \mathcal{N}\left(\frac{(\sigma^2 + \rho\sigma\tau)z}{\sigma^2 + 2\rho\sigma\tau + \tau^2}, \frac{(1-\rho^2)\sigma^2\tau^2}{\sigma^2 + 2\rho\sigma\tau + \tau^2}\right)$$

$$\Rightarrow (c) E[X|X+Y=z] = \frac{(\sigma^2 + \rho\sigma\tau)z}{\sigma^2 + 2\rho\sigma\tau + \tau^2}, \quad (d) \text{var}[X|X+Y=z] = \frac{(1-\rho^2)\sigma^2\tau^2}{\sigma^2 + 2\rho\sigma\tau + \tau^2}$$